EXPERIMENT 28 : Implementation of Medical Diagnosis System Using Prolog

Aim:

To write and execute a Prolog program to implement a simple medical diagnosis system using symptoms and rules.

Software Required:

- SWI-Prolog
- Windows Operating System

Theory:

A medical diagnosis system is an expert system that uses predefined facts and rules to identify possible diseases based on symptoms.

In Prolog, facts represent symptoms, and rules are used to determine possible conditions. The system asks the user to enter symptoms and provides a possible diagnosis based on the entered information.

Program:

Save the file as exp28.pl in C:/AI Prolog/.

```
% Medical Diagnosis System

% Disease diagnosis rules

diagnosis(cold) :-
    symptom(runny_nose),
    symptom(sneezing),
    symptom(sore_throat).

diagnosis(flu) :-
    symptom(fever),
    symptom(headache),
    symptom(body_pain).
```

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diagnosis(malaria) :-
    symptom(fever),
    symptom(chills),
    symptom(sweating).

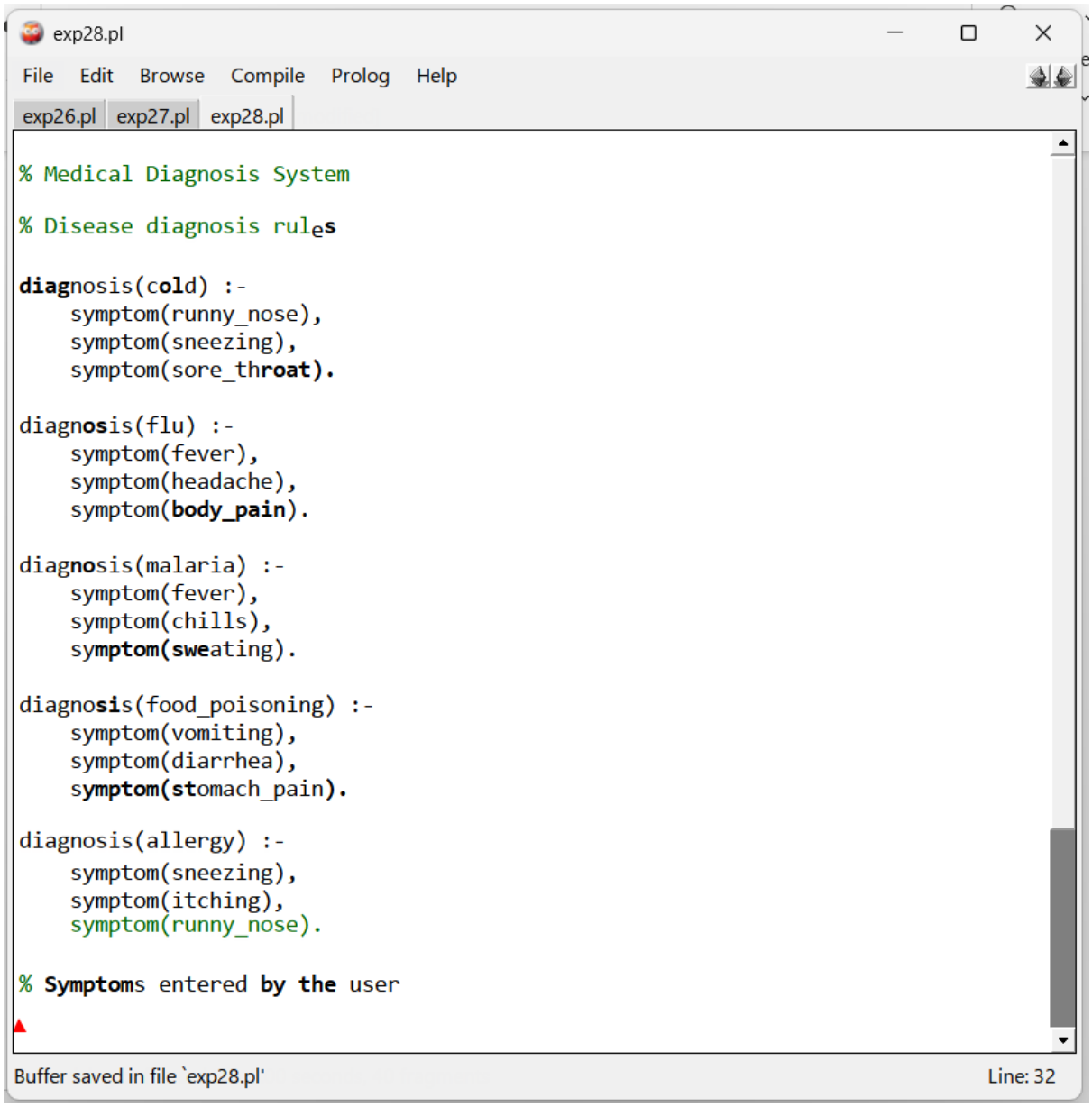
diagnosis(food_poisoning) :-
    symptom(vomiting),
    symptom(diarrhea),
    symptom(stomach_pain).

diagnosis(allergy) :-
    symptom(sneezing),
    symptom(itching),
    symptom(runny_nose).

% Symptoms entered by the user

symptom(fever).
symptom(headache).
symptom(body_pain).

% Display diagnosis
start :-
    diagnosis(Disease),
    write('Possible Disease: '),
    write(Disease),
    nl.
```



Procedure:

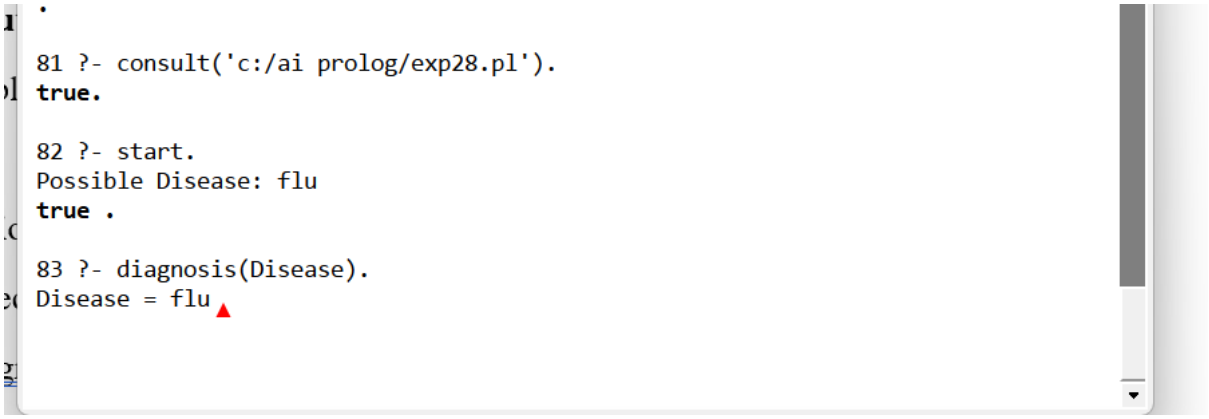
1. Open SWI-Prolog.
2. Create a new file named exp28.pl.
3. Type the above Prolog program.
4. Save the file in C:/AI Prolog/.
5. Load the program using the following command:

?- consult('c:/ai prolog/exp28.pl').

6. Execute the following query:

?- start.

Output:



Algorithm:

1. Define the diseases and their associated symptoms.
2. Store symptoms as facts in the knowledge base.
3. Apply diagnosis rules using Prolog.
4. Match the available symptoms with the rules.
5. Display the possible disease.

Result:

Thus, the Prolog program to implement a simple Medical Diagnosis System using facts and rules was successfully written and executed.